Lecture 6 revision sheet

Numerical solution of ODEs: time-stepping methods for initial value problems

This sheet lists the definitions, formulas and results from Lecture 6, without derivations or discussion. These are in the full notes (`ode-time-stepping-1.ipynb` and the homework notebook); the grey pointers give the section. → **L***n* marks material that later lectures build on. [*] marks optional and [**] advanced material in the notes.

Key messages

1. Time-stepping replaces $\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y})$ by a recurrence for $\boldsymbol{y}_n \approx \boldsymbol{y}(t_n)$. **Explicit** schemes compute $\boldsymbol{y}_{n+1}$ directly; **implicit** schemes need an (algebraic) equation solve each step, but usually have much better stability.
2. Three error measures: the **local truncation error** (LTE, error per unit step), the **local** (one-step) error $= \Delta t \times$LTE, and the **global** error at fixed time. For a stable method the global error has the same order p as the LTE.
3. **Consistency + stability $\iff$ convergence.** Consistency alone is not enough.
4. Stability is analysed on the test problem $y' = \lambda y$, with λ the eigenvalues of the (diagonalised) system, which may be complex. A scheme is (absolutely) stable when its **amplification factor** satisfies $|\sigma(\lambda \Delta t)| < 1$.
5. Whether a given Δt is stable depends on both the scheme *and* the physics (λ).
6. Runge–Kutta methods reach high order by evaluating $\boldsymbol{f}$ at intermediate stages. Classical RK4 (four evaluations, fourth order) is the popular "sweet spot".

1 Initial value problems

Full notes: *Initial Value Problems (IVPs); Requirements for a numerical scheme*

- **IVP:** find $\boldsymbol{y}(t)$ with $\dfrac{d\boldsymbol{y}}{dt} = \boldsymbol{f}(t, \boldsymbol{y})$ and $\boldsymbol{y}(0) = \boldsymbol{y}_0$. The solution is a trajectory in *phase space*; the ODE gives its local direction.
- **Higher-order ODEs** become first-order systems by introducing new variables, e.g. $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = 0$ with $v = \dot{x}$ gives $\dot{x} = v$, $\dot{v} = -2\zeta\omega_0 v - \omega_0^2 x$.
- **Notation:** time levels $t_0 < t_1 < \cdots$ with step $\Delta t = t_{n+1} - t_n$ (or a local Δt_n), and $\boldsymbol{y}_n \approx \boldsymbol{y}(t_n)$. Time level is a subscript here (a superscript, u_i^n, once space is discretised too).
- **Requirements:** (1) *convergence*, $\max_{0 \le t_n \le T} \|\boldsymbol{y}_n - \boldsymbol{y}(t_n)\| \to 0$ as $\Delta t \to 0$; (2) *correct qualitative behaviour* at finite Δt (e.g. boundedness, positivity, conservation, correct long-time limit).
- [*] Terminology: **autonomous** if $\boldsymbol{f} = \boldsymbol{f}(\boldsymbol{y})$ only; **linear** if $\boldsymbol{f}$ is linear in $\boldsymbol{y}$ ($y' = \sin t$ is linear, $y' = \sin y$ is not); **stiff** if stability, not accuracy, forces a small Δt (e.g. a small, fast component alongside the slow one of interest); **conserved quantities** that a good scheme should respect.

2 Model problems

Full notes: *Model Problem 1; Model Problem 2 [*]; homework*

Problem	Exact solution	Notes
Relaxation: $y' = K(y_r - y)$	$y = y_r + (y_0 - y_r)e^{-Kt}$	$y \to y_r$. Eigenvalue $\lambda = -K$ (the constant Ky_r does not affect stability). Radioactive decay ($y_r = 0$): $K = \ln 2 / t_{1/2}$.
Linear test problem: $y' = \lambda y$	$y = y_0 e^{\lambda t}$	As $t \to \infty$: $y \to \infty$ if $\lambda > 0$, $y \to 0$ if $\lambda < 0$, $y \equiv y_0$ if $\lambda = 0$.
MSD system [*]: $\boldsymbol{y} = (x, v)^T$, $\boldsymbol{y}' = A\boldsymbol{y}$ with $A = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & -2\zeta\omega_0 \end{pmatrix}$	see below	Mass–spring–damper: $\omega_0 = \sqrt{k/m}$ (undamped frequency), $\zeta = c/\sqrt{4mk}$ (damping ratio).
Circular orbit [*]: $\dot{x} = -y$, $\dot{y} = x$	$(x, y) = (\cos t, \sin t)$	From $(1, 0)$; $r = \sqrt{x^2 + y^2}$ is conserved. Eigenvalues $\pm i$.
Lorenz system	none (chaotic)	$\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$, with $\sigma = 10$, $\rho = 28$, $\beta = 8/3$.

MSD eigenvalues [*]: $\det(A-\lambda I)=\lambda^2+2\zeta\omega_0\lambda+\omega_0^2=0\Rightarrow\lambda_{1,2}=\omega_0\big(-\zeta\pm\sqrt{\zeta^2-1}\big)$.

$\zeta=0$	purely imaginary $\lambda=\pm i\omega_0$	undamped (simple harmonic motion)
$0<\zeta<1$	complex, $\mathrm{Re}(\lambda)=-\zeta\omega_0<0$	under-damped
$\zeta\geq 1$	real, negative	over-damped ($\zeta=1$: critically damped)

Under-damped exact solution, with $\omega=\omega_0\sqrt{1-\zeta^2}$: $x(t)=e^{-\zeta\omega_0 t}\Big(x_0\cos\omega t+\dfrac{\zeta\omega_0 x_0+v_0}{\omega}\sin\omega t\Big)$.

3 Four simple schemes

Full notes: *Link to numerical differentiation; Four simple schemes; An example application* → **used in L7**

Replace dy/dt at a time level by a finite difference: *forward* $\frac{y_{n+1}-y_n}{\Delta t}$, *backward* $\frac{y_n-y_{n-1}}{\Delta t}$ or *central* $\frac{y_{n+1}-y_{n-1}}{2\Delta t}$. Writing $f_n=f(t_n,y_n)$:

Scheme	Update	Type	Order	Amplification factor $\sigma(z)$, $z=\lambda\Delta t$
Forward (explicit) Euler	$y_{n+1}=y_n+\Delta t\,f_n$	explicit, 1-step	1	$1+z$
Backward (implicit) Euler	$y_{n+1}=y_n+\Delta t\,f_{n+1}$	implicit, 1-step	1	$1/(1-z)$
Trapezoidal	$y_{n+1}=y_n+\frac{\Delta t}{2}(f_n+f_{n+1})$	implicit, 1-step	2	$(1+z/2)/(1-z/2)$
Leapfrog	$y_{n+1}=y_{n-1}+2\Delta t\,f_n$	explicit, 2-step	2	roots of $\sigma^2-2z\sigma-1=0$

- **Implicit** schemes give an equation for y_{n+1} (nonlinear and/or a matrix system in general). For the scalar linear relaxation problem with $y_r=0$ they can be rearranged: BE $y_{n+1}=y_n/(1+K\Delta t)$; trapezoidal $y_{n+1}=y_n(1-K\Delta t/2)/(1+K\Delta t/2)$.
- Trapezoidal = average of forward and backward Euler, which cancels their leading errors.
- **Leapfrog is not self-starting** (it needs y_1 as well as y_0): start with one step of a self-starting method, e.g. forward Euler. Its *spurious mode* causes the oscillations seen on the decay problem. It needs only one f evaluation per step (popular in weather prediction, with filters).
- On the decay problem: FE and BE are both first order with almost identical asymptotic errors; leapfrog and trapezoidal are second order, with leapfrog's errors larger because of its oscillations.

```
for n in range(len(t)-1):  y[n+1] = y[n] + dt*f(t[n], y[n])
```
(forward Euler, generic RHS `f(t, y)`)

4 Errors, consistency and convergence

Full notes: *Errors and stability; Definitions (errors); Error analysis for forward Euler* → **used in L7**

Definitions

- **Local truncation error (LTE)** τ: substitute the *exact* solution into the scheme, written in the form that approximates the derivative ($\frac{y_{n+1}-y_n}{\Delta t}=\ldots$, not $y_{n+1}=y_n+\Delta t\times\ldots$; other texts may differ by one power of Δt). For forward Euler

$$\tau=\frac{y(t_{n+1})-y(t_n)}{\Delta t}-f(t_n,y(t_n))=\frac{\Delta t}{2}y_n''+\frac{\Delta t^2}{6}y_n'''+\mathcal{O}(\Delta t^3).$$

- **Order of accuracy** p: $|\tau|\leq C\Delta t^p$ as $\Delta t\to 0$, i.e. $\tau=\mathcal{O}(\Delta t^p)$, with p the largest such power. **Consistent**: $\tau\to 0$ as $\Delta t\to 0$ (i.e. $p>0$).
- **Local (one-step) error**: the error made in one step starting from exact data; local error $=\Delta t\,\tau=\mathcal{O}(\Delta t^{p+1})$. Forward Euler: $\frac{1}{2}\Delta t^2 y_n''=\frac{1}{2}\Delta t^2\frac{df}{dt}$, so it grows with Δt and with how fast f changes.
- **Global error**: $E=\max_{t_0\leq t_n\leq T}\|\boldsymbol{y}_n-\boldsymbol{y}(t_n)\|$ at a *fixed* final time T. Convergence means $E\to 0$ as $\Delta t\to 0$.

- **Consistency + stability $\iff$ convergence.** For a stable method of order p: LTE $\mathcal{O}(\Delta t^p)$, local error $\mathcal{O}(\Delta t^{p+1})$, global error $\mathcal{O}(\Delta t^p)$ (about $T/\Delta t$ steps accumulate the local errors).
- [*] The stability in that statement is *zero-stability*: errors stay bounded on a fixed $[0,T]$ as $\Delta t\to 0$ (Dahlquist; the PDE analogue is the Lax equivalence theorem). All the methods here are zero-stable, so all converge. §5 is about *absolute stability*: does the solution of $y'=\lambda y$ decay for a *fixed* Δt as $t\to\infty$?
- Measure the global error at a *fixed time*. With a fixed *number of steps* instead, the observed rate moves towards the local-error rate.
- **LTE vs residual:** the LTE puts the exact solution into the discrete scheme (estimated by Taylor series); a *residual* puts the numerical solution into the continuous equation (computable).

- Orders are measured as slopes of log–log error plots (L4).

5 Stability

Full notes: *Continuous problem analysis; Stability of the continuous problem; Numerical stability* → **used in L7**

Continuous problem

- Systems $\boldsymbol{y}' = A\boldsymbol{y}$ decouple by **diagonalisation** [*]: $A = S\Gamma S^{-1}$ (columns of S the eigenvectors, $\Gamma = \mathrm{diag}(\lambda_i)$), $\boldsymbol{y} = S\boldsymbol{z}$ gives $z_i' = \lambda_i z_i$. So it suffices to study the scalar test problem $z' = \lambda z$ for each eigenvalue, which can be complex even when A is real (L3). (At critical damping, $\zeta = 1$, the MSD matrix cannot be diagonalised, but the stability conclusions still hold.)
- $z' = \lambda z$, $z(0) = 1$ is **stable** ($z \to 0$, the fixed point $z = 0$ attracts the solution) iff $\mathrm{Re}(\lambda) < 0$, and **unstable** ($|z| \to \infty$) iff $\mathrm{Re}(\lambda) > 0$, since $|e^{\lambda t}| = e^{\mathrm{Re}(\lambda)t}$ (Euler's formula: the imaginary part only oscillates). If $\mathrm{Re}(\lambda) = 0$, $|z(t)| = 1$ for all t (e.g. the undamped MSD).

Discretised problem

- A one-step scheme applied to $z' = \lambda z$ gives $z_{n+1} = \sigma(\lambda\Delta t)\, z_n$, so $z_n = \sigma^n z_0$. It is **stable** iff $|\sigma(\lambda\Delta t)| < 1$ for every eigenvalue. The **stability region** is the set of $\lambda\Delta t$ in the complex plane where this holds (axes $\mathrm{Re}(\lambda)\Delta t$, $\mathrm{Im}(\lambda)\Delta t$).

Scheme	Stability region (where $\lvert\sigma\rvert < 1$)
Continuous problem	left half-plane $\mathrm{Re}(\lambda) < 0$ (no Δt)
Forward Euler	disc $\lvert 1 + \lambda\Delta t\rvert < 1$: centre -1, radius 1. Conditionally stable; never stable for purely imaginary λ.
Backward Euler	outside the disc $\lvert 1 - \lambda\Delta t\rvert \le 1$ (centre $+1$, radius 1). Contains the whole left half-plane: **unconditionally stable** (A-stable). Also "stable" for some $\mathrm{Re}(\lambda) > 0$, where the exact solution grows: e.g. $\lambda\Delta t = 3$ gives $\sigma = -\frac{1}{2}$ per step against the exact factor $e^3 \approx 20$ (stable but qualitatively wrong).
Trapezoidal	exactly the left half-plane $\mathrm{Re}(\lambda\Delta t) < 0$: unconditionally stable (A-stable).
Leapfrog	only the imaginary-axis segment $\lambda\Delta t = iy$, $\lvert y\rvert < 1$, where $\lvert\sigma\rvert = 1$ (bounded, no growth). Elsewhere the spurious mode, $\sigma \approx -(1 - \lambda\Delta t)$, grows (for real $\lambda < 0$ it flips sign every step with $\lvert\sigma\rvert > 1$).

Applied to the model problems

- **Relaxation** ($\lambda = -K$), forward Euler: $|1 - K\Delta t| < 1 \iff 0 < K\Delta t < 2 \iff \Delta t < 2/K$. The solution is also monotonic only if $K\Delta t \le 1$. With $K = 0.04$ the limit is $\Delta t < 50$:

Δt	8	16	32	64
$1 - K\Delta t$	0.68	0.36	-0.28	-1.56
Behaviour	monotonic	monotonic	stable, wiggles early on	unstable

- **MSD** [*], forward Euler, $0 \le \zeta < 1$: $|1+\lambda\Delta t| < 1 \iff \Delta t < 2\zeta/\omega_0$ (L3). Examples ($\omega_0 = 1$): $\Delta t = 0.04, \zeta = 0.1$ stable; $\Delta t = 0.2, \zeta = 0.05$ unstable; $\Delta t = 0.2, \zeta = 0.2$ stable. Undamped ($\zeta = 0$): unstable for every Δt, so a different scheme is needed.
- The stable Δt is set by the physics: roughly inversely proportional to the fastest time scale ($|\lambda|$).

6 Runge–Kutta methods

Full notes: *Runge-Kutta (RK) methods* → **used in L7**

RK methods are single-step, *multi-stage* methods: they evaluate f at intermediate points (*stages*) within the step, rather than using earlier time levels (multistep methods).

Method	Stages	Order
RK2: improved Euler (Heun, explicit trapezoidal)	$k_1 = f(t_n, y_n),\ k_2 = f(t_n + \Delta t,\ y_n + \Delta t\, k_1)$, $y_{n+1} = y_n + \frac{\Delta t}{2}(k_1 + k_2)$	2
RK2: explicit midpoint	$k_1 = f(t_n, y_n),\ k_2 = f(t_n + \frac{\Delta t}{2},\ y_n + \frac{\Delta t}{2} k_1)$, $y_{n+1} = y_n + \Delta t\, k_2$	2
Classical RK4	$k_1 = f(t_n, y_n),\ k_2 = f(t_n + \frac{\Delta t}{2},\ y_n + \frac{\Delta t}{2} k_1)$, $k_3 = f(t_n + \frac{\Delta t}{2},\ y_n + \frac{\Delta t}{2} k_2)$, $k_4 = f(t_n + \Delta t,\ y_n + \Delta t\, k_3)$, $y_{n+1} = y_n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$	4

- Improved Euler is a *predictor–corrector*: a forward Euler guess y^* replaces y_{n+1} in the implicit trapezoidal rule. Explicit midpoint does the same with the implicit midpoint rule. (Names vary: "modified Euler" is used for both RK2 methods.)
- [**] General RK2(α): $k_2 = f(t_n + \alpha\Delta t,\ y_n + \alpha\Delta t\, k_1)$, $y_{n+1} = y_n + (1 - \frac{1}{2\alpha})\Delta t\, k_1 + \frac{1}{2\alpha}\Delta t\, k_2$; $\alpha = 1$ is improved Euler, $\alpha = \frac{1}{2}$ explicit midpoint.
- Example ($y' = y + t^3$, $y(0) = 1$, one step $\Delta t = 1$; exact $y(1) = 3.028$): improved Euler 3.000, explicit midpoint 2.625, RK4 3.010.
- RK4 achieves order 4 with 4 function evaluations; orders above 4 need more stages than their order (e.g. order 5 needs 6), which is why RK4 is so popular.

RK stability [**]: on $y' = \lambda y$ the amplification factors are truncated exponential series.

Method	$\sigma(z)$, $z = \lambda\Delta t$	Stability
RK2 (any α)	$1 + z + \frac{z^2}{2}$	Real axis: $-2 < \lambda\Delta t < 0$ (as FE). Larger region for complex λ, but never stable on the imaginary axis ($\lvert\sigma\rvert^2 = 1 + y^4/4$).
RK4	$1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$	Real axis: $-2.79 < \lambda\Delta t < 0$. Imaginary axis: $\lambda\Delta t = iy$, $0 < \lvert y\rvert < 2\sqrt{2} \approx 2.83$. Also small patches with $\mathrm{Re}(\lambda) > 0$.

7 Results from the homework

Full notes: *homework notebook*

- On $y' = y$ (error at $t = 3$): log–log slopes ≈ 1 (forward Euler), 2 (improved Euler), 4 (RK4). SciPy's `odeint` gives much smaller errors: it adapts its step size (and order) to meet an error tolerance.
- **Circular orbit:** forward Euler spirals outwards, since $\lvert 1 \pm i\Delta t\rvert = \sqrt{1 + \Delta t^2} > 1$ multiplies r every step; improved Euler stays much closer to the circle ($\lvert\sigma\rvert = \sqrt{1 + \Delta t^4/4}$). Check conserved quantities (r; energy H and angular momentum L for Kepler orbits [**]) to judge qualitative behaviour.
- **Lorenz system:** trajectories from different solvers stay close at first, then diverge (chaos); the difference stays bounded because the trajectories stay on the attractor.